

Exact norm density of the fixed atomic algebra

MathlibAnnex.CStarAlgebra.CAR.hasDensityCharacter_atomicCounterexampleAlgebra

theorem

Combines a cardinality bound from a faithful representation on a separable Hilbert space with a lower bound for every norm-dense subset.

Statement

Let A be the completed CAR algebra and $T = C^*(\pi(A) \cup \{L_i : i \in I\}) \subseteq B(\mathcal{H})$ the algebra generated by the selected direct sum π of pure-state GNS representations indexed by their unitary-equivalence classes $i \in I$ and the unitaries L_i from the fixed homogeneity and shell construction. Write $\iota(a) = \pi(a)$ for the embedding $A \rightarrow T$. Write $\mathfrak{c} = 2^{\aleph_0}$. The norm-density character of T is exactly $\mathfrak{c}$: there is a norm-dense subset $D_0 \subseteq T$ with $|D_0| = \mathfrak{c}$, and every norm-dense subset $D \subseteq T$ satisfies

$$\mathfrak{c} \leq |D|.$$

The density character is the least cardinality of a norm-dense subset.

Assumptions

The fixed algebra T is infinite dimensional, has exactly one unitary-equivalence class of nonzero irreducible representations, and has a faithful representation $\tilde{\rho}$ on the separable space $H_\tau = L^2(A, \tau)$. These are established properties of this T , not additional hypotheses on an arbitrary algebra.

Conclusion

Thus $\text{dens}_{\|\cdot\|}(T) = \mathfrak{c}$. Separately, the underlying set also has $|T| = \mathfrak{c}$. In particular T is not norm separable, although it is faithfully represented on H_τ .

The dense-set lower bound is a statement about every norm-dense subset, not merely about the whole carrier. Neither cardinality is a Hamel dimension or a Hilbert-space dimension.

Proof route

The faithful representation on H_τ gives an upper bound for $|T|$. The cited density theorem for an infinite-dimensional unital C^* -algebra with a unique irreducible representation class gives a lower bound for every dense subset. Applying it also to T itself proves that this bound is attained.

▼ Proof steps

1. Injectivity of $\tilde{\rho} : T \rightarrow B(H_\tau)$ and separability of H_τ give

$$|T| \leq |B(H_\tau)| \leq \mathfrak{c}.$$

This is the cited cardinality bound for an algebra faithfully represented on a separable Hilbert space.

2. Let $\rho_0 : T \rightarrow B(\mathcal{H})$ be the inclusion representation and let $D \subseteq T$ be norm dense. The cited lower-bound theorem applies to the irreducible inclusion ρ_0 , the fact that every nonzero irreducible unital representation is equivalent to it, and infinite dimensionality of T . It gives $\mathfrak{c} \leq |D|$. Its argument is by contradiction: $|D| < \mathfrak{c}$ would give a dense set of size less than $\mathfrak{c}$ in an irreducible representation space, and the finite-dimensionality theorem at that density would force T to be finite dimensional.
3. Taking $D = T$ gives $\mathfrak{c} \leq |T| \leq \mathfrak{c}$, hence $|T| = \mathfrak{c}$. The choice $D_0 = T$ then attains the lower bound, proving the exact density statement.

Main citations

- The stated existence or structural result —
MathlibAnnex.CStarAlgebra.CAR.hasDensityCharacter_atomicCounterexampleAlgebra
- Cardinality bound from a faithful representation on a separable space —
MathlibAnnex.CStarAlgebra.CAR.cardinalMk_atomicCounterexampleAlgebra_le_continuum
- The lower bound for every dense subset of this algebra —
MathlibAnnex.CStarAlgebra.CAR.continuum_le_cardinalMk_dense_atomicCounterexampleAlgebra
- The separate carrier-cardinality equality —
MathlibAnnex.CStarAlgebra.CAR.cardinalMk_atomicCounterexampleAlgebra
- The density bound for a single irreducible representation class —
MathlibAnnex.Analysis.CStarAlgebra.Representation.continuum_le_cardinalMk_dense_algebra_of_singleton_of_not_finiteDimensional
- Faithful separable representations bound the carrier —
MathlibAnnex.Analysis.CStarAlgebra.Representation.cardinalMk_le_continuum_of_injective

Supporting route explanation

Lean source signature (exact)

```
theorem hasDensityCharacter_atomicCounterexampleAlgebra :
  HasDensityCharacter AtomicCounterexampleAlgebra Cardinal.continuum
```

In the source	Mathematical meaning
AtomicCounterexampleAlgebra	The one fixed CAR shell-generated algebra T , equipped with its operator-norm topology.
Cardinal.continuum	The continuum cardinal $\mathfrak{c} = 2^{\aleph_0}$, rather than a linear or Hilbert-space dimension.
HasDensityCharacter AtomicCounterexampleAlgebra Cardinal.continuum	The full conclusion has two clauses: there is a norm-dense subset $D_0 \subseteq T$ with $ D_0 = \mathfrak{c}$; and every norm-dense $D \subseteq T$ satisfies $\mathfrak{c} \leq D $. Thus $\text{dens}_{\ \cdot\ }(T) = \mathfrak{c}$. The separate equality $ T = \mathfrak{c}$ mentioned in the text is not substituted for this minimality assertion.

In the source	Mathematical meaning
Dense s ; $\#s$	In the linked exact definition of <code>HasDensityCharacter</code> , <code>Dense s</code> means norm density of the subset here and $\#s$ is its cardinality, not its vector-space dimension.

Further source notes: In the linked proof, the first supplier bounds `#AtomicCounterexampleAlgebra`; the second bounds `#s` for each dense set s . The `#` notation is cardinality, not dimension.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.hasDensityCharacter_atomicCounterexampleAlgebra`

Accepted content SHA-256: `b40286a5ff6686d5c043943f269bb42e40bbe62442dae691b3332697d8177473`

Accepted source guide SHA-256: `c9883867d594fea0ac7aa0919806e625452543972c701d32bb858d343b9d75a6`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`